

VORTEX-GEN PITCH DECK

1. The Next-Gen Aerodynamic Platform

The Problem

- Overly Complex: Traditional CFD software requires hours of setup and complex meshing.
- Computationally Expensive: Needs dedicated hardware to run a single simulation.
- Inaccessible: Extremely unfriendly to beginners with massive licensing fees.

The Solution: Vortex-Gen

Vortex-Gen democratizes aerodynamic analysis. We've built a full-stack, production-grade simulation platform that combines interactive WebGL visualizations with a powerful PyTorch Neural Network backend to deliver real-time fluid dynamics predictions.

- Instant Feedback: Millisecond-level C_l/C_d predictions using our NeuralFoil AI.
- Interactive WebGL Engine: Beautiful, real-time particle flows and CFD smoke.
- Collaborative Workspaces: Secure team environments for rigorous A/B comparisons.

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2. Business Model & Features

Core Capabilities

- The Wind Tunnel Lab: Drop in any airfoil and instantly visualize mathematically accurate Von Kármán vortex streets and live telemetry.
- AI-Powered Autotune: Stop guessing. Let our Deep Tune AI exhaustively sweep thousands of NACA coordinates to find the absolute maximum Lift-to-Drag ratio.
- Professional Export Artifacts: Instantly extract optimized geometries as 3D-printable STLs or multi-page branded analytical PDF reports.

SaaS Business Model

Vortex-Gen operates on a tiered SaaS subscription model seamlessly integrated with Stripe:

- Free Tier: Basic simulation, local analytic model, and 1 custom import per month.
- Pro Tier: Unlocks the PyTorch solver, Fast Tune Optimizer, and 10 imports/month.
- Pro Max Tier: Unlimited imports, Deep Tune Optimizer, and Team Workspaces.